New Palaeoproterozoic palaeomagnetic data from the Kaapvaal Craton, South Africa

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Abstract: Palaeomagnetic data from the well-dated 2060.6 ± 0.5 Ma Phalaborwa Complex in South Africa (Kaapvaal Craton) are of excellent quality. High unblocking components are carried by magnetite and single polarity remanence directions (mean declination 5.0°, inclination 57.3°, $\alpha_{95} = 5.2°$) yield a palaeomagnetic pole (latitude 27.7°N, longitude 35.8°E, A95 = 6.6°) that overlaps with existing poles from the near coeval 2054.4 ± 1.3 Ma Bushveld Complex. The Phalaborwa and Bushveld complex poles, along with poles from the well-dated Vredefort impact (2023 ± 4 Ma) and Post-Waterberg Dolerites (1874.6 ± 3.9 Ma), define the most reliable poles for the Kaapvaal Craton during this time interval (*c.* 2060–1875 Ma) and witness low rates of Mid-Palaeoproterozoic apparent polar wander. Poorly dated NE–NNE-trending dyke swarms that intrude the Phalaborwa and Bushveld complexes both yield dual-polarity remanence components that share a common mean at the 95% confidence level. Primary palaeomagnetic poles (Phalaborwa dykes pole latitude 7.6°, longitude 12.1°, A95 = 11.8°; Bushveld dykes pole latitude 12.6°, longitude 24.1°, A95 = 10.8°) suggest that they are of the same age as the Post-Waterberg dolerites (*c.* 1875 Ma). They could also be as old as the Phalaborwa and Bushveld Complexes, however; high-precision geochronology is required to resolve this issue and to enlarge the number of Palaeoproterozoic key poles for the Kaapvaal Craton.

Pangea, the youngest and best-documented supercontinent on Earth, formed at the end of the Palaeozoic era. In the deep past, however, Neoproterozoic (Rodinia), Palaeoproterozoic (Columbia) and Archaean (Vaalbara) supercontinents/continents have been postulated (e.g. Hoffman 1991; Dalziel 1992; Torsvik *et al.* 1996; Cheney 1996; Zegers *et al.* 1998; Rogers & Santosh 2002; Zhao *et al.* 2002; Meert & Torsvik 2003; Torsvik 2003; Li *et al.* 2008; de Kock *et al.* 2009; Evans 2009). However, the timing for assembly/dispersal and continental geometry of these older supercontinents/continents differs in almost every paper. The Archaean-aged Vaalbara continent, hypothetically composed of the Kaapvaal (South Africa) and Pilbara (West Australia) cratons, is no exception and we show three different constellations (Fig. 1a) that can be seen in the literature. Vaalbara may have existed between 2.8 and 2.1 Ga (Cheney 1996; Zegers *et al.* 1998; Strik *et al.* 2003; de Kock *et al.* 2009) but Zhao *et al.* (2002) argue that Vaalbara was also intact at 2.0–1.8 Ga and became the site of two major orogenic events. The first event is that of the Limpopo mobile belt (2.0–1.9 Ga), which formed as a result of the collision between the Kaapvaal and Zimbabwe cratons (Zeh *et al.* 2009). The second orogenic event resulted in the formation of the Capricorn Orogen, in which the Pilbara and Yilgarn cratons were sutured together. These orogens, together with many other 2.1–1.8 Ga orogens (Zhao *et al.* 2002), were argued to be the result of a world-wide amalgamation of cratonic blocks to form the Columbia supercontinent (Fig. 1b).

In order to test the hypothesis of the Columbia supercontinent, high-precision geochronological and palaeomagnetic data from many Palaeoproterozoic cratons are urgently required. In this account we report new palaeomagnetic data from the *c.* 2.06 Ga Phalaborwa Complex and dykes cutting both the Phalaborwa and Bushveld (*c.* 2.05 Ga) complexes in South Africa, and summarize the most

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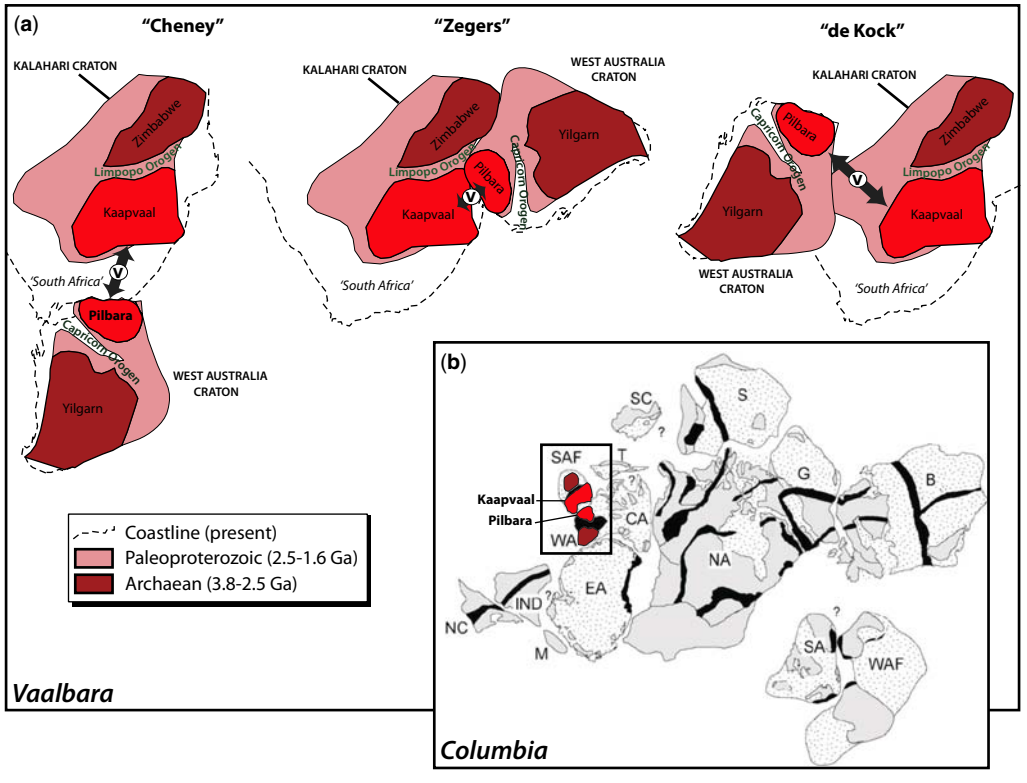


Fig. 1. (**a**) Vaalbara continent showing the proposed Archaean connection between the Kaapvaal (South Africa) and Pilbara (West Australia) cratons according to Cheney (1996), Zegers *et al.* (1998) or de Kock *et al.* (2009). See de Kock *et al.* (2009) for reconstruction parameters for a fixed South Africa (Kaapvaal). Archaean and Palaeoproterozoic cratons that make up the basement of Kalahari and West Australia follow Gubanov & Mooney (2009). (**b**) Reconstruction of the Columbia Supercontinent (Zhao *et al.* 2002), mostly based on correlation of 2.1–1.8 Ga orogens (B: Baltica; CA: Central Australia; EA: East Antarctica; G: Greenland; IND: India; M: Madagascar; NA: North America; NC: North China; S: Siberia; SA: South America; SAF: South Africa (Kalahari); SC: South China; T: Tarim; WA: West Australia; WAF: West Africa). In this reconstruction, the Vaalbara continent is intact at Columbia time and shown in a Cheney (1996) type fit.

reliable palaeomagnetic poles (*c.* 2.06–1.85 Ga) from the Kaapvaal/Kalahari Craton. There are no reliable 2.0 Ga poles from the Pilbara Craton (or Australia at large), and hence proposed connections between the Kaapvaal and Pilbara cratons (Vaalbara continent) remain unproven at this time. A comprehensive global analysis and aspects of the Columbia supercontinent will be dealt with elsewhere.

Phalaborwa and Bushveld

The Phalaborwa Complex (Harmer 2000) outcrops in an elongated kidney shape in the Limpopo Province in northern South Africa (Fig. 2). The Complex was intruded into Archaean basement gneisses at 2060.6 $\pm$ 0.5 Ma (U–Pb baddeleyite; Reischmann 1995) in several alkaline cycles, which emplaced a suite of rocks ranging from ultramafic to peralkaline in character. The first phase, the intrusion of a massive ultramafic pipe-like body, consists mainly of pyroxenites. This was followed by an alkaline phase in which the surrounding syenite plug-like bodies were emplaced. At a later stage, foskorite (olivine + magnetite + apatite + phlogopite rock) and banded carbonatite of the central pegmatoid body were emplaced. The final phase saw the intrusive injection of the transgressive carbonatite (PMC 1976). The Phalaborwa Complex is nearly coeval with the Bushveld Igneous Complex (2054.4 $\pm$ 1.3 Ma; Scoates & Friedman 2008), the world's largest layered intrusion.

Regional aeromagnetic data reveal that northern South Africa is riddled with dykes (Fig. 3). Unfortunately, very little is known about the age of those occurring in the Phalaborwa and Bushveld

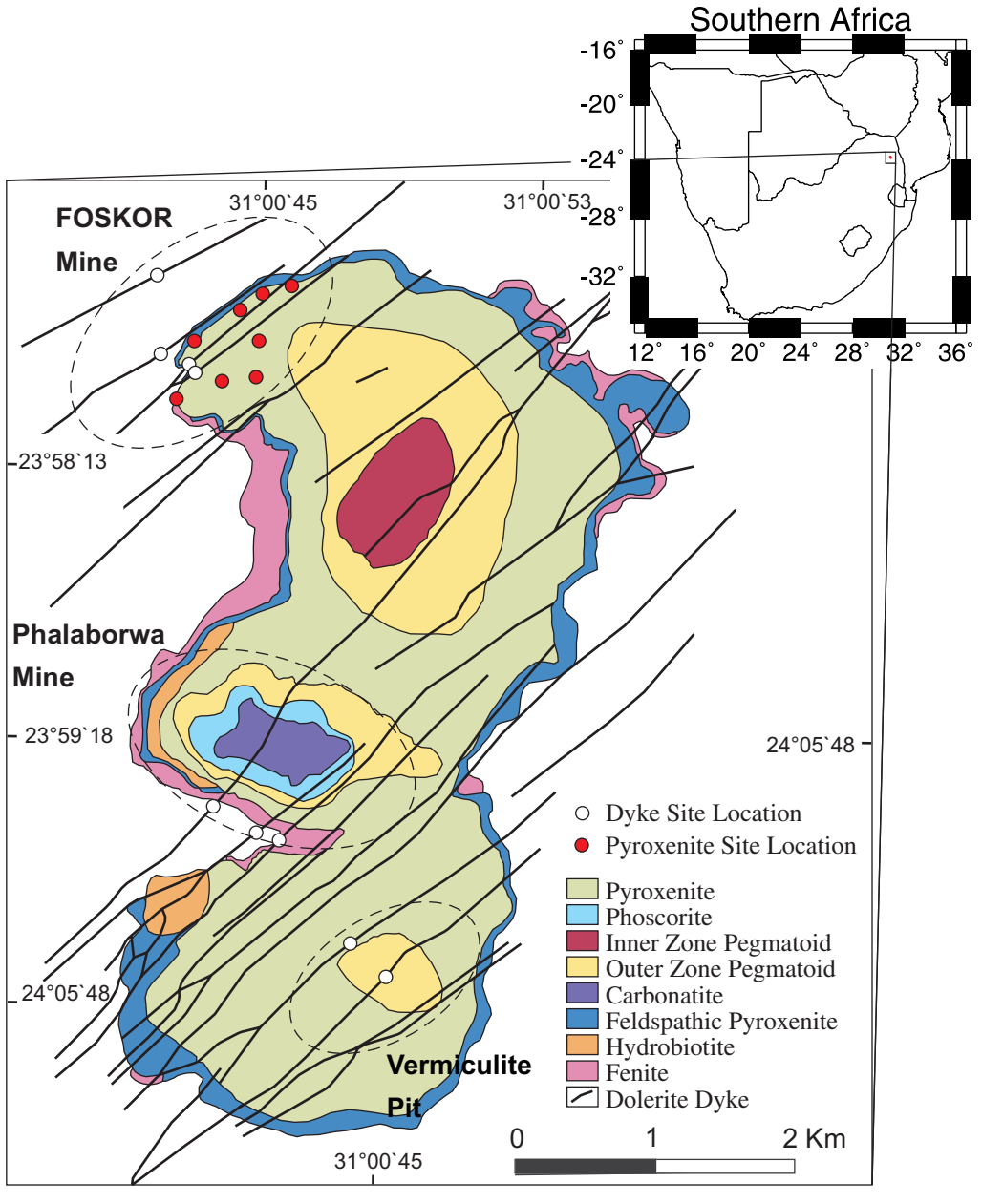


Fig. 2. Simplified geological map of the Phalaborwa Complex with site locations obtained from the pyroxenites (red circles) and dykes (open white circles). The three pits/mines are shown by dashed outline modified from Morgan & Briden (1981).

Complexes. Examination of dyke swarms and their relationship to major mafic magmatic events in the Kaapvaal Craton reveals that dyke swarms have repeatedly exploited certain orientations (Uken & Watkeys 1997). Three main trends have been identified in which dyke swarms were repeatedly emplaced: NE, NW and east–west. Ages and orientations of major dyke swarms for the entirety of southern Africa are summarized by Hunter & Reid (1987). In this study, only the NE-trending swarm that intrudes the Phalaborwa and Bushveld complexes are considered (Figs 2–4).

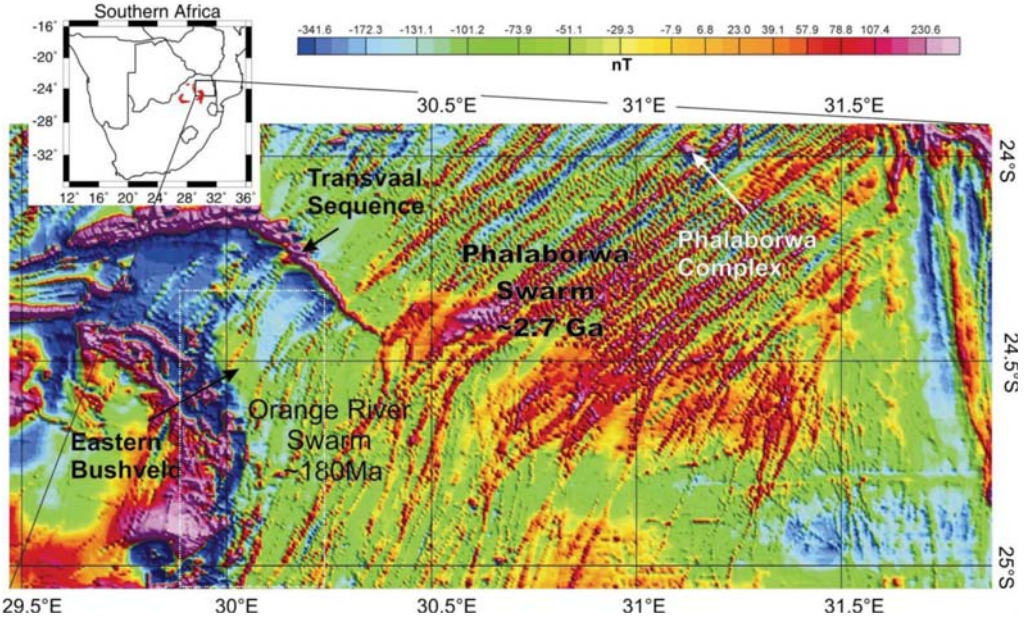


Fig. 3. Regional aeromagnetic data over northern South Africa (provided by the South African Council for Geoscience). The major geological bodies in the area are visible: Bushveld Complex, Phalaborwa Complex, Transvaal Sequence, as well as the Orange River and Phalaborwa dyke swarm. White stippled box represents the area depicted by Figure 4.

A mafic dyke database from the International Institute for Geo-information Science and Earth Observation (Chavez Gomez 2001) suggests that the dykes present in the Bushveld Complex are only from the Orange River Swarm (Fig. 3), that is, Jurassic in age (Karoo, *c.* 180 Ma). However, a palaeomagnetic study conducted on a few of these dykes (Fig. 4) revealed that the palaeomagnetic pole position clearly does not resemble any Karoo pole but instead indicates a Palaeoproterozoic age (Letts *et al.* 2005 and this study).

Sampling and laboratory procedures

All samples were obtained with the use of a portable drill and orientated via sun and magnetic compasses. The natural remanent magnetization (NRM) was measured on a JR6A magnetometer at the Geological Survey of Norway (Trondheim). NRM stability was tested by both thermal (MM-TD-60 furnace) and alternating field (AF; 3-axis *in-house* built tumbler) demagnetization. Bulk susceptibility was measured on a Bartington MS2 system. Thermomagnetic analysis (TMA) and determination of Curie temperatures were carried out on a horizontal translation bridge, while petrographic analysis was undertaken to identify the opaque mineralogy visually. Characteristic remanence components were calculated with the LineFind algorithm (Kent *et al.* 1983) as implemented in the Super-IAPD software (Torsvik *et al.* 2000).

Palaeomagnetic results

Phalaborwa Complex (pyroxenites)

The main Phalaborwa mine (Fig. 2) was unsafe to enter due to sidewall instability, while in other areas exposure was inadequate for palaeomagnetic sampling. It was therefore only possible to sample eight pyroxenite sites in the northern Foskor pit; a total of 90 core samples were analysed for NRM and bulk susceptibility (Table 1).

Figure 5 shows a representative selection of orthogonal vector plots that demonstrate the demagnetization behaviour. Many samples contain two

Fig. 4. The position of dyke samples collected from the Eastern Bushveld Complex: four sites from this study (yellow-Hackney mine and blue-Modikwa mine) and sites (white dots) from the author's previous study (Letts *et al.* 2005), plotted on high-resolution aeromagnetic data draped on top of regional aeromagnetic data courtesy of Anglo Platinum. Coordinates are in WGS84 (LO 31). See Figure 3 for geographic location.

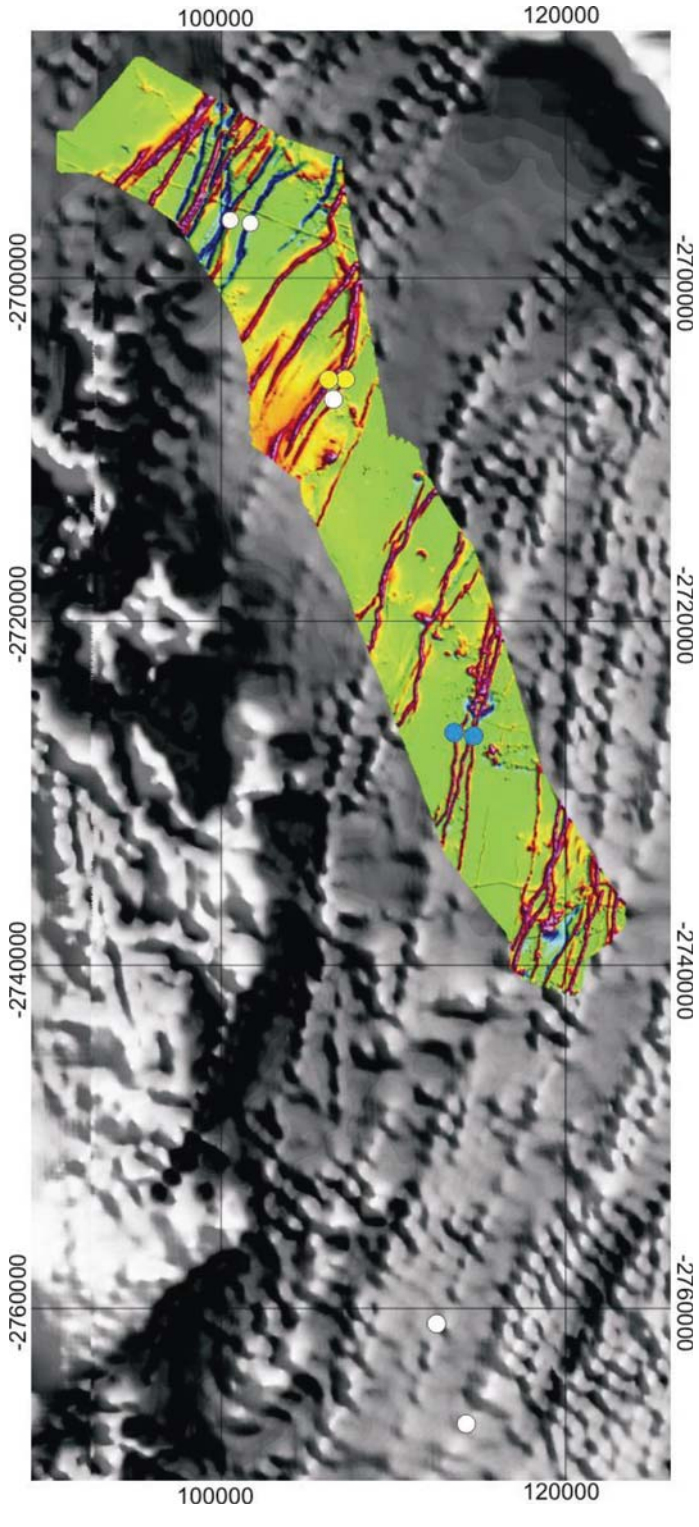


Fig. 4.

Table 1. *Mean NRM and bulk susceptibility for sites from (a) Phalaborwa Complex pyroxenite, (b) Phalaborwa dykes and (c) Bushveld dykes*

Site	NRM Decl.	NRM Incl.	α_{95}	N	Intensity average (A/M)	$\pm$ (A/m)	Susc. average ($\times 10^{-6}$ SI)	$\pm$ ($\times 10^{-6}$)	Q ratio
(a) Phalaborwa Complex pyroxenite									
P-1	353.4	48.3	7.9	9	214.5	17.1	226.0	42.0	
P-2	1.4	28.1	30.2	13	64.6	37.6	237.3	56.9	
P-3	356.9	33.1	29.2	9	17.1	4.7	327.3	6.4	
P-4	15.5	41.7	2.9	11	74.7	23.5	248.1	25.1	
P-5	6	34.9	4.7	13	3.5	1.5	228.0	30.4	
P-6	352.8	54.2	50.2	8	12.6	0.8	255.6	89.6	
P-7	13.1	19.2	7.6	15	115.3	37.1	4531.7	1277.6	
P-8	35.4	35.6	14.8	12	14.10	9.9	213.9	18.5	
(b) Phalaborwa dykes (F = FOSKOR pit, P = Phalaborwa pit, V = Vermiculite pit)									
F-1	11.8	50.9	6.3	7	3.15	3.8	28750.2	351.8	4.5
F-2	121.3	−63.9	1.3	10	8.48	3.2	33458.3	628.0	10.6
F-3	129.6	−68.7	3.6	10	8.76	11.7	35131.99	3779.7	10.4
F-4	158.7	−66.8	5.3	10	6.79	19.0	29177.5	837.4	9.7
P-1	167.1	−78.9	3.4	10	15.70	5.7	49172.5	772.5	13.3
P-2	317.3	69.7	20	5	30.83	20.4	15907.4	42325	81.2
P-3	20.6	35.1	73.9	11	1.38	3.3	40557.1	473.2	1.4
V-1	328.3	4.6	29.4	9	2.29	28.5	39453.6	1131.04	2.4
V-2	354.6	54.9	49.6	7	4.00	7.4	21442.0	1169.4	7.8
(c) Bushveld dykes (M = Modikwa Mine, H = Hackney Mine, B = Letts *et al.* (2005).									
M-1	170.4	−62.8	4.6	9	10.96	2130.7	36942	662.6	12.4
M-2	178.3	−63.5	4	10	12.47	13204.1	35449	1896.5	14.7
H-2	186.7	−67.9	6.1	10	3.35	529.4	23159	225.3	6
H-1	211.6	−65.4	12.4	7	3.27	253.7	25422	750	5.3
B-1	212	−73	6.2	8	3.387	1225	17288	9502	9.6
B-2	138	−71	3.8	9	4.282	447	30184	1507	6.4
B-3	6	−79	19.1	5	0.69	31	10473	5988	3.6
B-4	343	84	19.1	3	0.43	18	24154	1523	0.7
B-5	12	65	13.6	6	22.1	1041	27114	6297	36.8

Decl: declination, Incl: inclination, α_{95}: 95% confidence circle around mean direction, N: number of samples, $\pm$: 1-σ error.

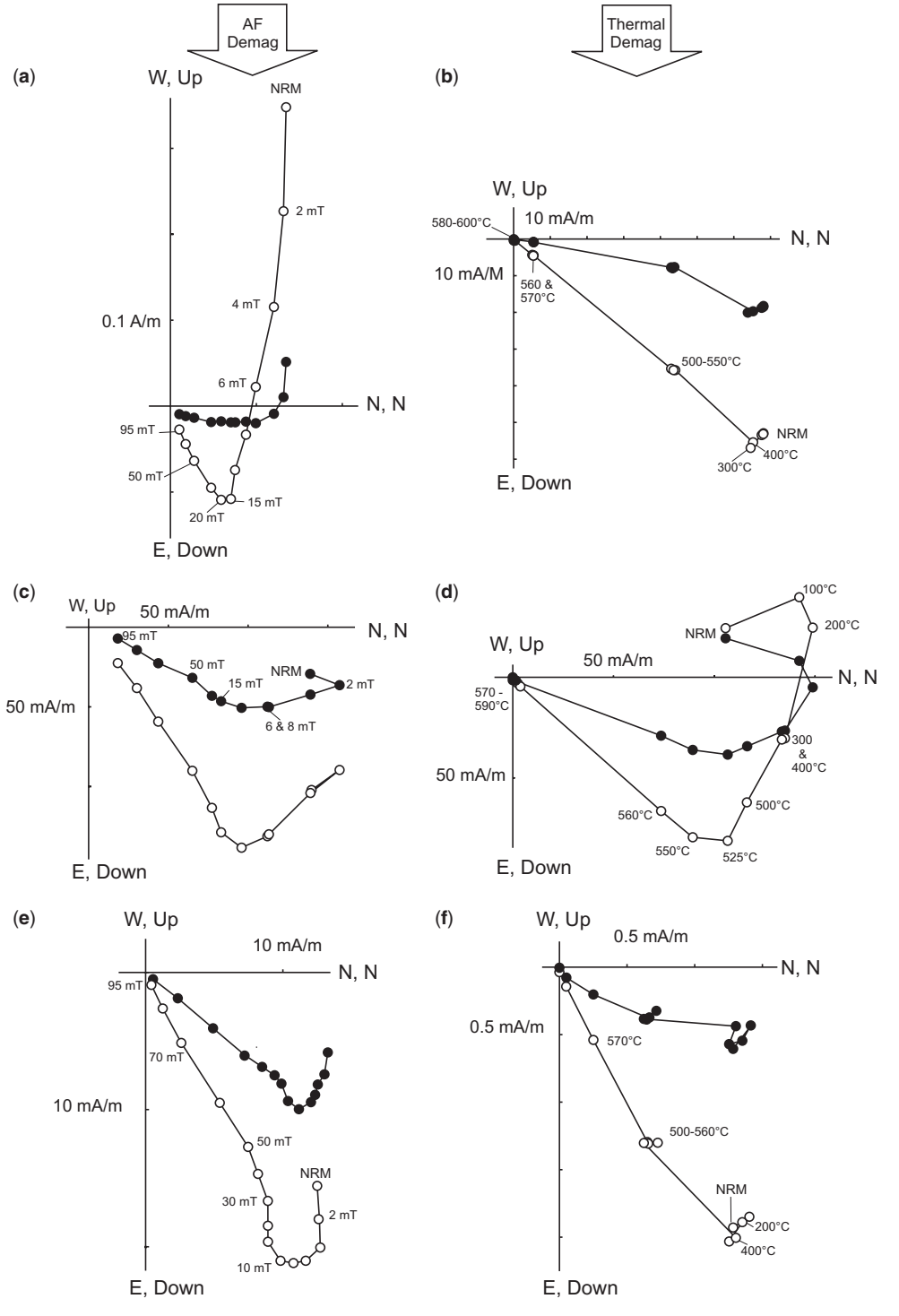


Fig. 5. Characteristic examples of AF (**a**, **c**, **d**) and thermal (**b**, **d**, **e**) demagnetization of samples (sites P-7 and P-8) from the Phalaborwa Complex (pyroxenite). In orthogonal vector plots, open (solid) symbols represent projections onto the vertical (horizontal) planes.

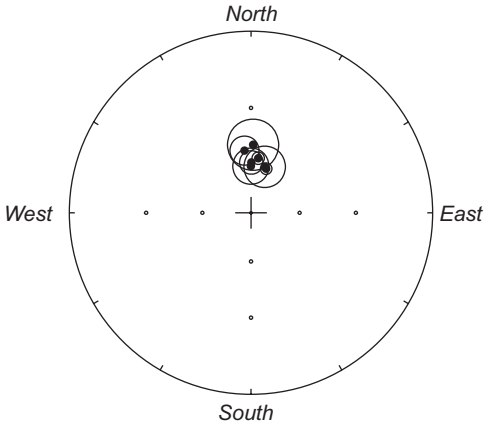


Fig. 6. Site mean directions and α_{95} confidence circles for Phalaborwa pyroxenite samples. Closed symbols denote downwards-pointing (positive) inclinations.

components of magnetization: a low coercivity or low unblocking (LB) component and a well-defined high-stability or high-temperature blocking (HB) component. HB components are typically identified above 500–550 °C or 20–50 mT, and thermal and AF demagnetization yielded similar directional results. HB components showed reasonable within-site grouping and site-mean directions are characterized by northerly declination and steep positive inclination (Fig. 6, Table 2). Samples from site P-6 behaved unstably during demagnetization and have not been included in the findings. LB components are scattered at site level and are not considered further.

Typical response to TMA is shown in Figure 7a, in which samples indicate a Curie temperature of *c.* 580 °C corresponding to magnetite. Petrographic examination discloses abundant sulphides, mostly pyrrhotite, but magnetite occurred as subordinate small discrete grains (Fig. 7b).

Phalaborwa dykes (dolerites)

Nine dolerite dykes were sampled in the Phalaborwa Complex: four in the Foskor pit, three in the main Phalaborwa Pit and two in the Vermiculite Pit (Fig. 2). A total of 79 dolerite dyke cores were analysed and NRM and bulk susceptibility values are listed in Table 1. None of the dykes have been dated.

Samples of the dykes behaved exceptionally well to both thermal and AF demagnetization, showing the presence of two components of magnetization (Fig. 8). LB components are random within sites and typically removed at around 300–500 °C or 10–30 mT. HB components show reasonable within-site grouping (Table 2). Site-mean directions plot with southerly to SE declination and very steep negative inclination or antipodal to this (Fig. 9). One site plots slightly off the other site-means but is included in the final statistics.

TMA (Fig. 7c, d) indicates Curie temperature at *c.* 580 °C. Analysis of samples from site V-2 revealed that, upon heating, the saturation magnetization first increased at 150–170 °C and then followed a normal decay for a Curie temperature at *c.* 580 °C. The kink at 150–170 °C, a typical low-temperature titanomagnetite (maghemitization) phenomenon (Ade-Hall *et al.* 1971), is probably related to hydrothermal alteration. Most samples are dominated by pure magnetite with a number of grains showing skeletal textures indicative of rapid cooling (Fig. 7e). A minor number of magnetite grains have oxy-exsolution lamellae of ilmenite (Fig. 7f). Samples from the V-2 site showed minor alteration of magnetite that could not be identified due to grain sizes being smaller then the optical resolution. Petrographic observations are in agreement with the findings obtained by thermomagnetic analysis, that is, that magnetite is the dominant carrier of magnetization.

Bushveld dykes

Samples from four underground dyke sites were obtained from the Eastern Bushveld Complex, two from the Hackney mine and two from the Modikwa mine. These sites are combined with five sites previously reported in Letts *et al.* (2005). The locations of all nine sites (Fig. 4) are plotted on high-resolution regional aeromagnetic data over the Bushveld Complex. The dykes are not adequately dated (see below).

Sixty-five core samples were analysed for NRM and bulk susceptibility (Table 1). Samples behaved well to both thermal and AF demagnetization, showing the presence of two components of magnetization (Fig. 10). LB components are typically removed below 500 °C or 30–40 mT. Samples obtained on the surface have stronger LB components relative to those obtained underground. Once again, LB directions are not related to any significant geological event and are therefore not considered. After demagnetization, one of two antipodal HB components is identified (Fig. 11). Most sites show directions with steep negative inclination and south to south-easterly declination. The remaining two sites show opposite polarity. Most sites (except B3, B4 and B5), showed good within-site grouping with α_{95} at around 3° (Fig. 11; Table 2).

TMA (Fig. 7g, h) indicates Curie temperature at *c.* 580 °C, once again suggesting magnetite or

Table 2. *Site mean HB components from (a) Phalaborwa Complex pyroxenite, (b) Phalaborwa dykes and (c) Bushveld dykes*

Site	Dec.	Inc.	α_{95}	N	κ	Lat. (S)	Lon. (E)
(a) Phalaborwa Complex pyroxenite							
P-1	359.8	61.2	10.7	9	74.96	23.962°	31.129°
P-2	1.8	48.9	14.1	13	43.34	23.962°	31.127°
P-3	354	52	8.2	9	127.82	23.969°	31.121°
P-4	7.5	56.3	3.6	11	454.69	23.966°	31.122°
P-5	0.3	59	6.9	13	122.73	23.968°	31.127°
P-7	19	61.2	3.4	15	168.04	23.965°	31.125°
P-8	16.5	60.4	12.2	12	30.61	23.963°	31.125°
Mean	*5.0*	*57.3*	*5.2*	*7*	*137.2*	*23.96*	*31.13*
Pole	Latitude 27.9°, Longitude 35.6° ($dp/dm = 5.6°/7.6°$); based on site directions						
	Latitude 27.7°, Longitude 35.8° (A95 = 6.6°); based on site poles						
(b) Phalaborwa dykes							
F-1	14.8	52.8	4.4	8	158.1	23.928°	31.096°
F-2	121	−62.9	1.7	9	959.5	23.965°	31.121°
F-3	132.3	−66.6	3	11	234.7	23.967°	31.122°
F-4	141.4	−66.9	3.2	10	230.2	23.967°	31.122°
P-1	168.1	−77.9	3.2	10	232.0	23.997°	31.125°
P-2	342.4	74.8	6.2	6	401.1	23.998°	31.128°
P-3	321.2	70.5	3.8	10	159.9	23.998°	31.128°
V-1	314.2	61.3	8	7	93.3	24.004°	31.135°
V-2	315.7	72.7	21	7	14.3	24.007°	31.136°
Mean	*328.0*	*69.2*	*7.9*	*9*	*43.6*	*23.99°*	*31.12°*
Pole	Latitude 8.3°, Longitude 12.2° ($dp/dm = 11.5°/13.5°$); based on site directions						
	Latitude 7.6°, Longitude 12.1° (A95 = 11.8°); based on site poles						
(c) Bushveld dykes							
H-1	185.4	−70.4	2.8	5	1885.7	24.350°	30.060°
H-2	180.5	−65.2	3.4	10	201.2	24.350°	30.060°
M-1	167.2	−64.3	2.6	9	401.6	24.183°	29.866°
M-2	172.9	−60.2	3.3	10	215.4	24.183°	29.866°
B-1	185	−73	3.2	8	298.54	24.462°	30.051°
B-2	137	−68	3.5	9	220.51	24.949°	30.151°
B-3	110.1	−75.9	20.1	5	21.84	24.883°	30.116°
B-4	359.9	62.2	18	3	48.12	24.413°	29.997°
B-5	10	65.6	12.8	6	28.19	24.370°	30.000°
Mean	*350.9*	*68.6*	*6.4*	*9*	*65.8*	*24.4°*	*30.0°*
Pole	Latitude 13.3°, Longitude 24.2° ($dp/dm = 9.1°/10.8°$); based on site directions						
	Latitude 12.6°, Longitude 24.1° (A95 = 10.8°); based on site poles						

N, number of samples; α_{95}, 95% confidence circle around the mean direction; κ, precision parameter; Lat. (S) and Lon. (E), mean sampling latitude (south) and longitude (east); A95, 95% confidence around the mean pole.

titanium-poor magnetite as the prime remanence carrier. Most samples showed almost reversible Curie curves upon cooling and heating, except for a small increase in saturation magnetization during cooling. This increase probably indicates secondary production of magnetite. TMA on a sample from site B-3 produced a distinctively different curve to those experienced in other sites. Upon heating, the saturation magnetization first increased at 150–170 °C followed by a distinctive decrease at *c.* 350 °C, suggesting low-temperature hydrothermal alteration (maghemitization) as also observed from some Phalaborwa dykes (Fig. 7d).

Petrographic analysis generally revealed quenched magnetite grains (indicating rapid cooling) as the dominant opaque mineralogy (Fig. 7i). A number of magnetite grains contain oxy-exsolution lamellae of ilmenite, possibly due to high-temperature deuteric oxidation during late-stage crystallization of the dyke (Fig. 7j). Samples from site B-3 are dominated by highly altered magnetite; the majority of the alteration mineral could

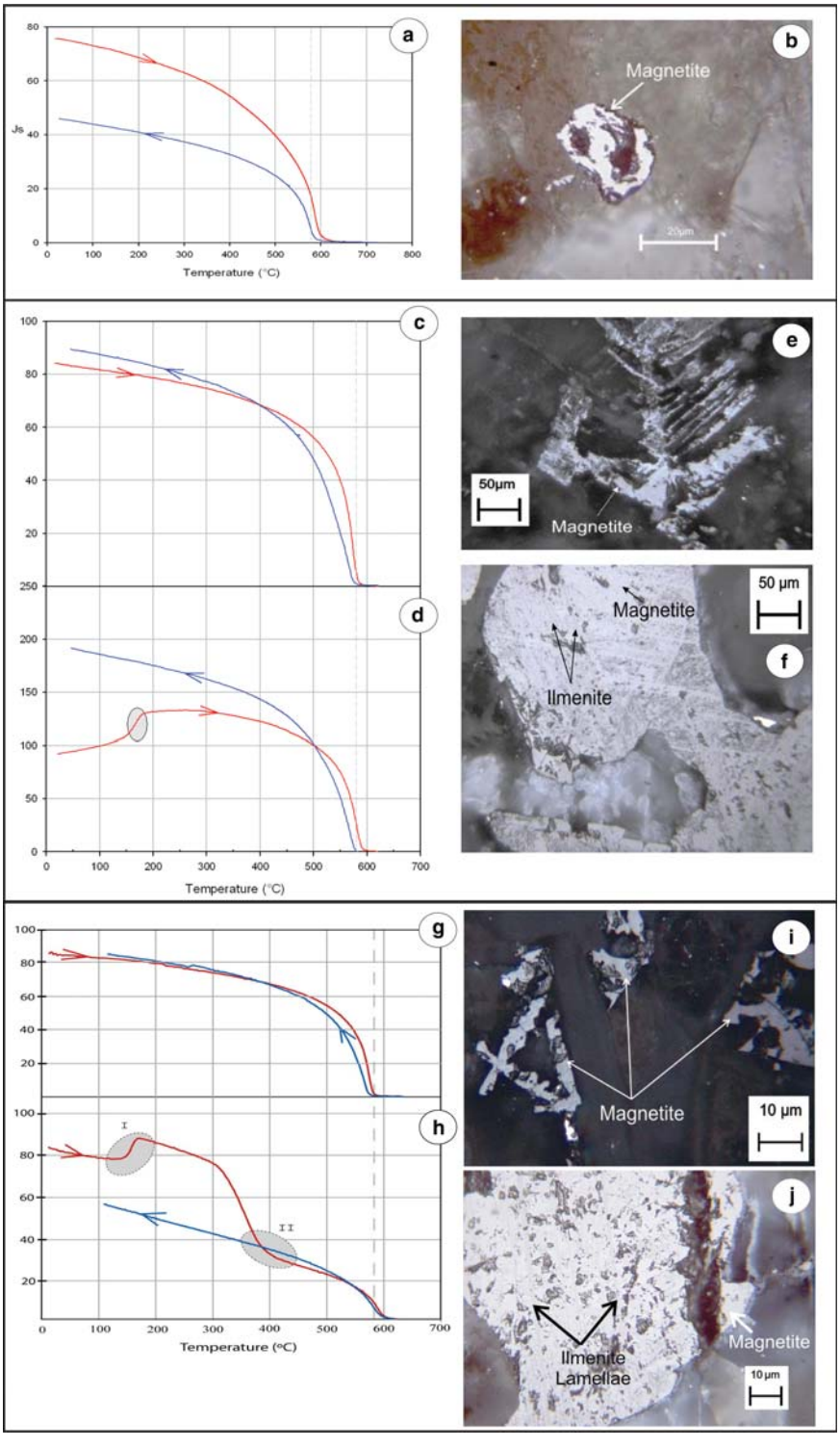

a
Js
Temperature (°C)
b
Magnetite
20µm
c
d
Temperature (°C)
e
50µm
Magnetite
f
50 µm
Magnetite
Ilmenite
g
h
I
II
Temperature (°C)
i
Magnetite
10 µm
j
Ilmenite
Lamellae
Magnetite
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Fig. 7.

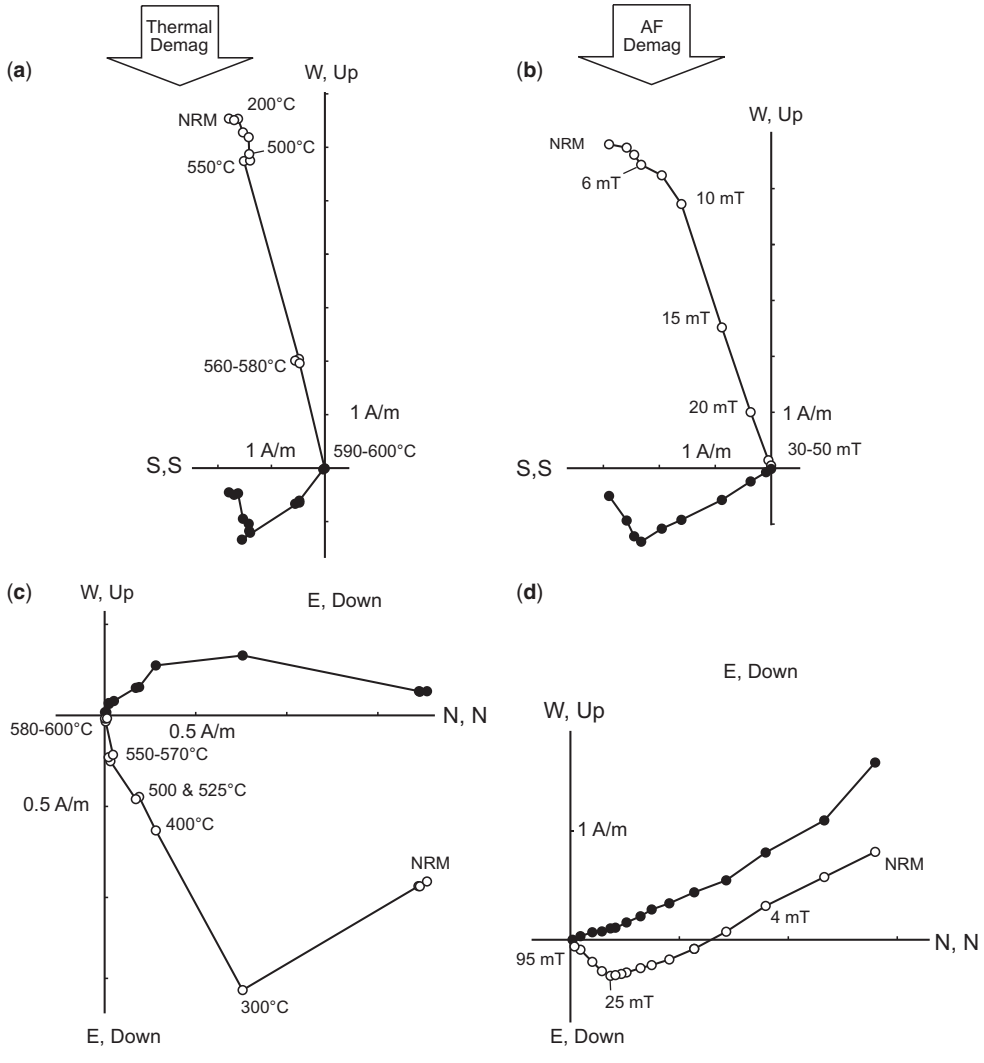


Fig. 8. Phalaborwa dykes: Typical orthogonal vector plots from thermal (**a**, **c**) and AF demagnetization (**b**, **d**). (a, b) are from Foskor site 4 (F-4) and (c, d) are from Vermiculite site 1 (V-1). Open (solid) symbols represent projections onto the vertical (horizontal) planes.

Fig. 7. Phalaborwa pyroxenite samples: (**a**) Typical thermomagnetic analysis from a sample from site P-4. Red line and arrow indicate heating phase while blue line and arrow indicates cooling phase. Curie temperature for magnetite at *c.* 580 °C observed (dashed line). (**b**) Discrete magnetite grain from site P-3 (reflected light). Phalaborwa dyke samples: (**c**, **d**) Thermomagnetic analysis of samples from sites F-3 and V-2. The majority of samples display Curie curves such as in (c), in which a small increase in saturation magnetization is seen during cooling, implying the possible production of more magnetite. In (d) there is irreversible creation of a magnetic phase with higher saturation magnetization at 150–180°, indicated by a grey ellipse. Curie temperature for magnetite is observed at *c.* 580 °C (dashed line). (**e**) Polished thin section showing a magnetite grain with a quenched texture from site F-3 (reflected light). (**f**) Polished thin section showing a magnetite grain with lamellae of ilmenite from site P-1. Bushveld dyke samples: (**g**, **h**) Thermomagnetic analysis of samples from sites M-1 and B-3. The majority of samples display Curie curves as in (h). In (i) there is irreversible creation of a magnetic phase with higher saturation magnetization at 150–180° (indicated by grey ellipse I) followed by inversion of maghemite to a weaker magnetic phase around 350 °C (indicated by grey ellipse II). (**i**) Polished thin section showing a magnetite grain with a quenched texture from site M-1 (reflected light). (**j**) Polished thin section showing a magnetite grain with lamellae of ilmenite, from site H-2.

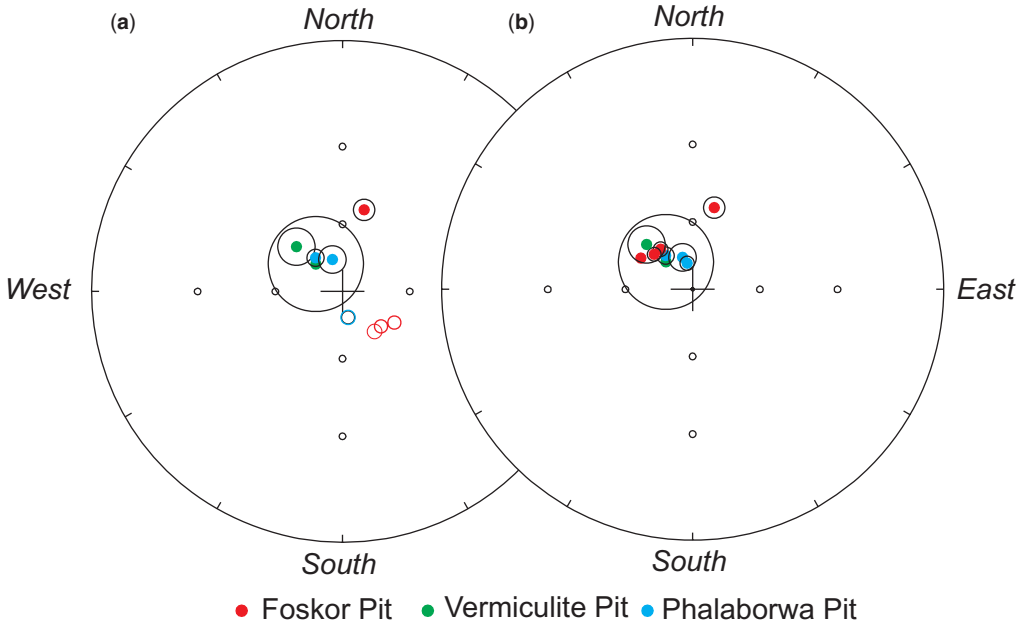


Fig. 9. (**a**) Phalaborwa dyke site-mean directions with α_{95} confidence circles and (**b**) sites inverted to positive inclination. Solid symbols indicate a positive inclination, while open symbols indicate a negative inclination. Note that α_{95} confidence circles for some sites are smaller than the symbol size.

not be determined due to grain sizes below the optical resolution. However, some alteration has been identified as hematite.

Interpretation

Phalaborwa Complex (pyroxenites)

The majority of samples contained minor LB components, successfully removed during demagnetization, and allowed for the identification of well-defined HB components that produce well-clustered results. Thermal demagnetization spectra and TMA suggest that Ti-poor magnetite or nearly pure magnetite (maximum unblocking temperatures of 565–580 °C and Curie temperatures of *c.* 580 °C) is the bulk remanence carrier. Petrographic analysis supports these observations.

Palaeomagnetic data reported in this study (Table 2) are in reasonable agreement with those of Morgan & Briden (1981) for the Phalaborwa Complex. A small observed difference in pole positions is probably due to blanket cleaning at *c.* 400 °C or 50 mT in the previous study. Our new pole for the Phalaborwa Complex pyroxenites (2060.6 ± 0.5 Ma) now plots closer to and statistically overlaps those obtained for the near coeval Bushveld Complex (2054.4 ± 1.8 Ma; Fig. 12)

Phalaborwa and Bushveld dykes

The age of the Phalaborwa and Bushveld dykes is uncertain and the Bushveld dykes have for long been assumed to be of Jurassic (Karoo) age. Palaeomagnetic data from both dyke sets are of excellent quality, and both normal and reversed polarities are recorded. The Phalaborwa dykes share a common mean at 95% confidence according to McFadden & Lowes (1981). The reversal test of McFadden & McElhinny (1990) also indicated a positive reversal test with a 'C' classification (i.e. the angle between the reversals is between 170° and 160°; critical angle 16.6°; observed angle 7.6°).

The Bushveld dykes also share a common mean at 95% confidence according to the method of McFadden & Lowes (1981). The McFadden & McElhinny (1990) reversal test produced a positive reversal test, with a 'C' classification (critical angle 15.1°; observed angle 9.3°). Positive reversal tests suggest that the reversely polarized dykes are of a similar age to the normally polarized dykes. Given the positive reversal tests, HB components isolated from both the Phalaborwa and Bushveld dykes were inverted to the same polarity before calculating mean site-directions and palaeomagnetic poles (Table 2). Given a statistical overlap in uncertainty levels from the two palaeomagnetic poles, the

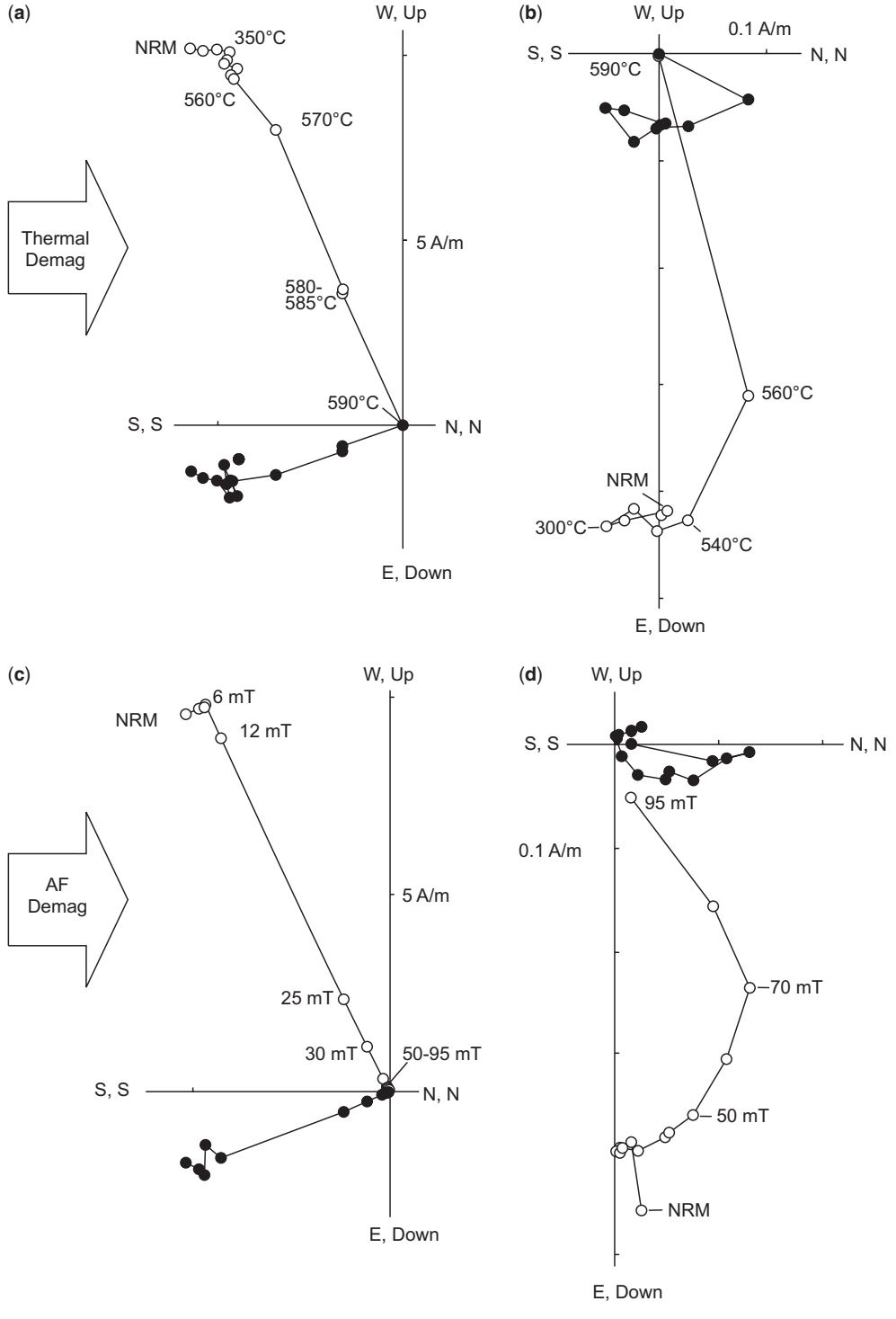


Fig. 10. Bushveld dykes: Typical orthogonal vector plots from thermal (**a**, **b**) and AF demagnetization (**c**, **d**). Samples obtained from underground are shown in (a, c). Surface samples obtained from Letts *et al.* (2005) are reflected in (b, d). Open (solid) symbols represent projections onto the vertical (horizontal) planes, respectively.

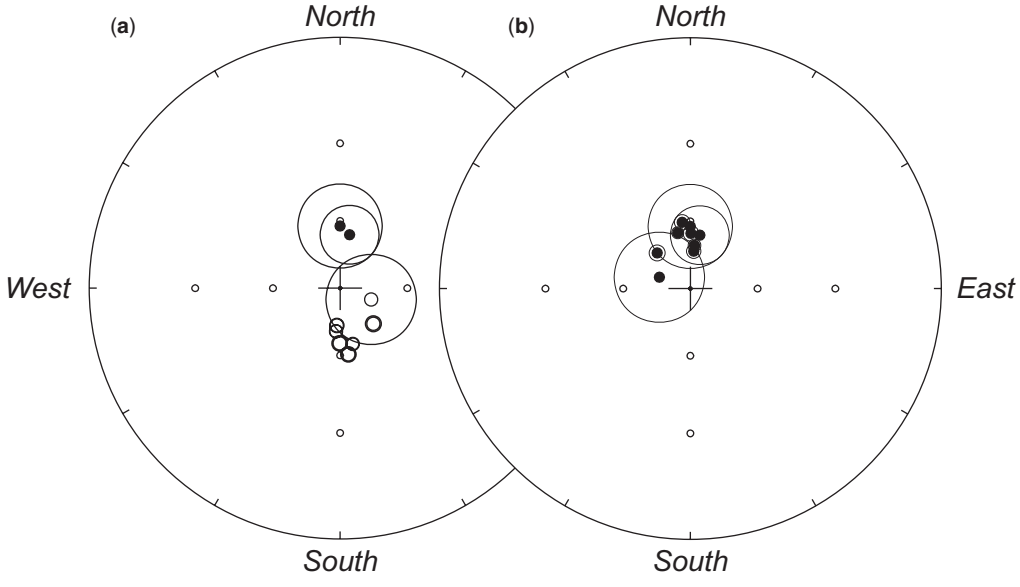


Fig. 11. Bushveld dyke site-mean directions (**a**) and sites inverted to positive inclinations (**b**). Solid symbols indicate a positive inclination, while open symbols indicate a negative inclination.

dykes could indeed be related to the same (but as yet undated) magmatic event.

Conclusions and key palaeomagnetic poles from the Kaapvaal/Kalahari Craton

Palaeomagnetic data from the well-dated Phalaborwa Complex (2060.6 $\pm$ 0.5 Ma) are of excellent quality, and HB components yield a palaeomagnetic pole (Fig. 12) that statistically overlaps with the near-contemporaneous Bushveld Complex (2054.4 $\pm$ 1.3 Ma). In the latter case normal and reversed directions share a common mean at the 95% confidence level (Letts *et al.* 2009), attesting to a primary magnetic signature. The Bushveld and Phalaborwa Complex poles must be regarded as the most reliable 2.05–2.06 Ga key poles for the Kaapvaal Craton.

Palaeomagnetic data from the dykes cutting both the Phalaborwa and Bushveld Complexes are also of excellent quality, and normal and reversed polarities from both dyke sets share a common mean at the 95% confidence level. The palaeomagnetic data clearly demonstrate that these dykes are Palaeoproterozoic and not Jurassic (Karoo) in age (Mean Karoo pole is 65.8°S and 83.7°E; A95 = 6.1; $N = 5$ poles; based on Torsvik *et al.* 2008). Compared with palaeomagnetic poles from southern Africa (Fig. 12a), both the Phalaborwa and Bushveld dyke poles statistically overlap with the well-dated (1874.6 $\pm$ 3.9 Ma; Hanson *et al.* 2004) post-Waterberg dolerites and the poorly dated (1878 $\pm$ 86 Ma; Barton 1979) Sand River Dykes (Morgan 1985). Based on pole positions, it is therefore possible that the dykes occurring in both the Phalaborwa and Bushveld complexes are the result of the same magmatic event at *c.* 1.9 Ga. However, they also plot close to the Bushveld Complex poles (Fig. 12a); an older age, but obviously $\leq$2.05–2.06 Ga (e.g. Bushveld age), cannot be precluded. An $^{40}Ar/^{39}Ar$ plagioclase age of 1649 $\pm$ 10 Ma was determined from one of the Bushveld dykes (Letts *et al.* 2005), but this age probably represents a low-temperature resetting event and not the emplacement age. Since the true emplacement age for these dykes is not established, the otherwise excellent primary palaeomagnetic poles from these dykes cannot be used as key poles before the dykes are properly dated; we are therefore awaiting geochronologists to elevate our palaeomagnetic poles as key Kaapvaal poles.

Many different Apparent Polar Wander (APW) paths have been published for the Kaapvaal Craton between *c.* 1.9 and 2.0 Ga. In Figure 12a we show five poles with blue confidence ovals that we consider as the only key poles for this time period. In addition to the Bushveld and Phalaborwa complexes and the Post-Waterberg Dolerites discussed above, we also include poles from the Vredefort Impact (2023 $\pm$ 4 Ma) and the Mashonaland Sill (1887.9 $\pm$ 1.2 Ma; weighted mean of ages reported

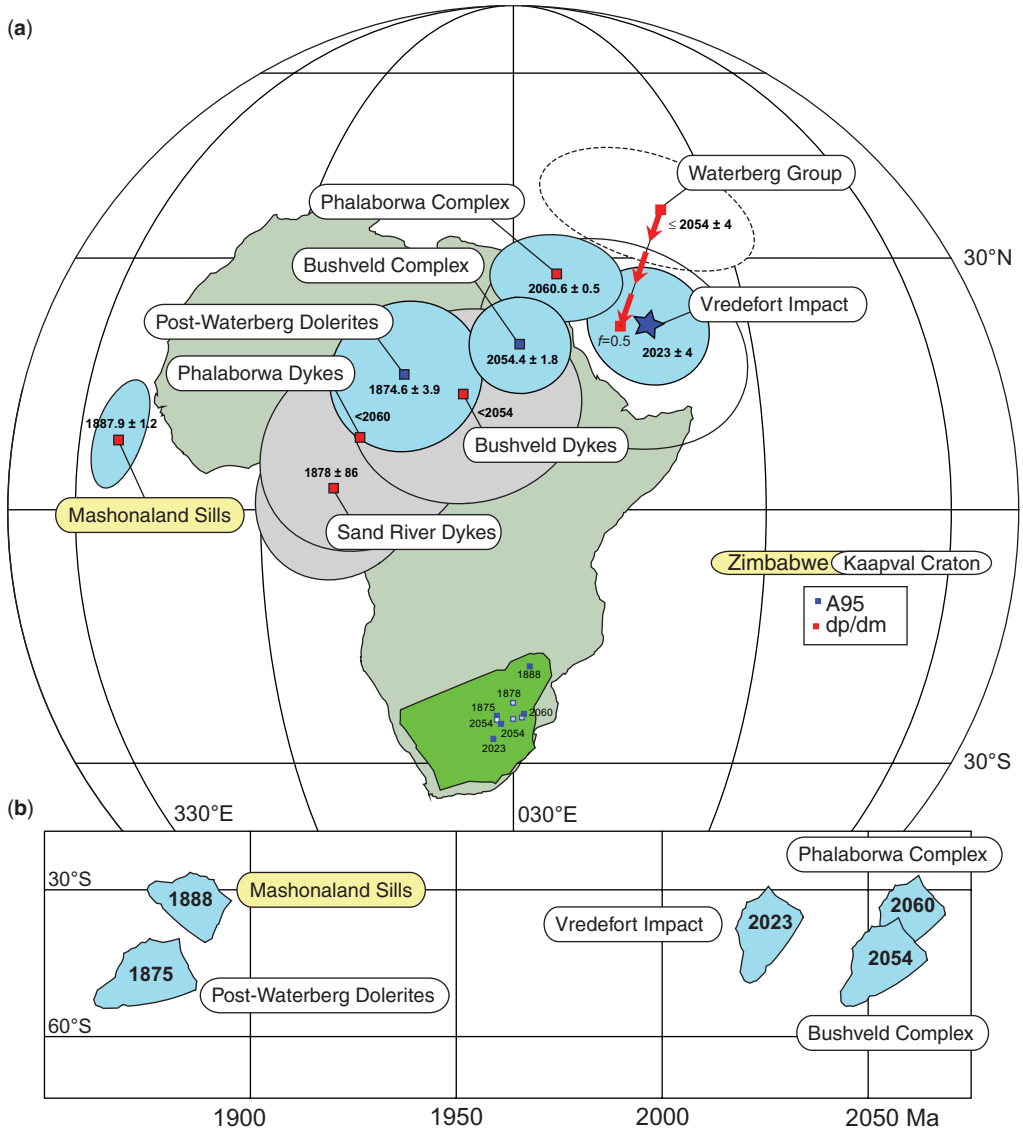


Fig. 12. (**a**) Selected palaeomagnetic poles from the Kaapvaal and Zimbabwe (only Mashonaland Sills) cratons from *c.* 2060 to 1875 Ma. Phalaborwa Complex pyroxenite, Phalaborwa dykes and Bushveld dykes are from this study (see text and Table 3 for other references). The outline of the Kalahari Craton (in green) and mean sampling positions for all poles are also shown. Note that some 95% confidence 'ovals' are reported as A95 circles (blue squares/star) whilst others are dp/dm ovals (red squares). The Waterberg Group pole has been corrected for sedimentary inclination shallowing using a flattening factor f of 0.5. Orthographic projection. (**b**) Reconstruction of the Kalahari Craton (latitude and orientation) according to poles in (a). Galls Projection.

in Söderlund *et al.* 2010). These are plotted along with less reliable poles obtained from the Waterberg Group ($\leq$2054 $\pm$ 4 Ma), the Bushveld and Phalaborwa dykes and the Sand River dykes (all three plotted with grey ovals/ellipses). The Sand River pole is not considered as a key pole due to large age uncertainties; all three dyke poles, however, plot near the well-dated Post-Waterberg Dolerites pole (1874.6 $\pm$ 3.9).

Of the nine poles shown in Figure 12a, of which five are reliable in terms of both palaeomagnetism and geochronology, two poses some concern.

Table 3. *The most reliable palaeomagnetic poles (2.06–1.87 Ga) from the Kaapval (notes 1–3, 5) and Zimbabwe (note 4) cratons in South Africa (blue ovals in Fig. 12a)*

Group/Lithology	Pole longitude	Pole latitude	A95 (dp/dm)	Age (Ma)	Notes
Phalaborwa Complex pyroxenite	27.7°N	35.8°E	6.6°	2,060.6 ± 0.5	1
Bushveld Complex	19.2°N	30.8°E	5.8°	2,054.4 ± 1.3	2
Vredefort Impact	21.8°N	44.5°E	(11.3/15.4°)	2,023.0 ± 4.0	3
Mashonaland Sills	7.6°N	338.2°E	5.1°	1,887.9 ± 1.2	4
Post-Waterberg Dolerites	15.6°N	17.1°E	8.9°	1,874.6 ± 3.9	5

Notes: (1) This study. (2) Five results obtained for each zone of the Bushveld Complex (Letts *et al.* 2009) have been grouped together to produce a single pole. (3) The 2023 ± 4 Ma (Kamo *et al.* 1996) Vredefort impact structure has been the subject of three palaeomagnetic studies. An initial study conducted by Hargraves (1970) sampled a range of lithologies within the central core and outer overturned collar rocks. A later study by Hart *et al.* (1995) obtained palaeomagnetic data from 16 sites within the central granitic basement of the structure. The most recent study (Carporzen 2006) concentrated on the pseudotachylites and granophyre dykes that are directly related to the impact event and obtained two palaeopoles: 19.9°N, 47.7°E and $dp/dm = 6.3°/8.6°$ for the pseudotachylites and 23.6°N, 41.3°E and $dp/dm = 3.6°/4.9°$ for the granophyre dykes. The pseudotachylites and granophyre results were grouped together to produce a single pole for the impact. (4) The well-dated Mashonaland Sills (Söderlund *et al.* 2010) occur as a suite of early Proterozoic intrusive dolerites that crop out in the north-eastern section of the Archaean Zimbabwe Craton. Here we use a recalculated palaeopole for the Mashonaland sills (Evans *et al.* 2002), in which they combined the dual-polarity results obtained by McElhinny & Opdyke (1964) and Bates & Jones (1996). (5) A number of dolerites occurring as sills, irregular bodies and dykes have intruded the Waterberg group in northern South Africa. Palaeomagnetic analysis conducted on samples located at or near localities from geochronological samples produced well-grouped antipodal directions (Hanson *et al.* 2004).

(i) The Waterberg Group pole (pole WUBS-I in de Kock *et al.* 2006) is 2.05 Ga or less in age but this pole is statistically different from both the Phalaborwa and Bushveld complexes and the *c.* 30 Ma younger Vredefort Impact pole (Fig. 12a).

(ii) The Mashonaland Sills pole is now well dated (1887.9 ± 1.2 Ma; Söderlund *et al.* 2010), the palaeomagnetic data (including a positive reversal test) appear robust (e.g. Bates & Jones 1996 and summarized in Evans *et al.* 2002), but the pole plots far away from the key Post-Waterberg Dolerite pole which is only 13.3 Ma (on average) younger.

The Waterberg Group pole is the only pole in our selection that is derived from sedimentary rocks. Inclination shallowing is a well-known phenomena in sedimentary rocks (e.g. Kent & Tauxe 2005), and by using a flattening factor f of 0.5 (a common/realistic value) we calculate a revised pole position for the Waterberg Group (WUBS-I) to 21.4°N and 43.2°E. This revised pole conforms much better with the near-coeval Phalaborwa and Bushveld Complex poles and the somewhat younger Vredefort Impact pole (Fig. 12a).

The Mashonaland Sills pole plots 39° of arc away from the Post-Waterberg Dolerite pole. This amounts to 3°/Ma, which is difficult to explain unless invoking true polar wander. Based on recent estimates of Earth properties, however, a true polar wander speed limit has been estimated as *c.* 1°/Ma (Steinberger & O'Connell 1997; Tsai & Stevenson 2007) although a more realistic numerical model based on subduction history suggests that up to 2°/Ma is possible for shorter time periods (Steinberger & Torsvik 2010). The Mashonaland Sills pole is the only pole that comes from the Zimbabwe Craton. An alternative reason for the pole discrepancy with the Post-Waterberg Dolerites pole is that the Zimbabwe and Kaapvaal cratons were not fully sutured at *c.* 1.9 Ga (as portrayed in Fig. 12b) with the Zimbabwe Craton located at lower latitudes (*c.* 30°S) than Kaapvaal, which was located to intermediate latitudes.

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